MIL-V-8712A
8 OCTOBER 1965

Superseding
MIL-V-8712(ASG)
9 August 1954

MILITARY SPECIFICATION

VALVE; AIR RELIEF, LOW PRESSURE

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers the requirements for one type of low-pressure relief valve for use in pressurized shipping containers.

1.2 Classification.- Air relief valves shall be one of the following sizes, as specified (see 6.2):

Size A - 3/8-inch external pipe thread pressure port
Size B - 1/2-inch external pipe thread pressure port

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

NN-P-515	Plywood, Container Grade
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Box, Fiberboard
PPP-T-60	Tape, Pressure Sensitive Adhesive, Waterproof - for Packaging and Sealing

Military

MIL-P-116	Preservation, Methods of
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-C-5501	Caps and Plugs, Protective, Dust and Moisture Seal
MIL-P-5516	Packings and Gaskets, Preformed, Petroleum Hydraulic Fluid Resistant
MIL-P-7105	Pipe Threads, Taper, Aeronautical National Form, Symbol ANPT, General Requirements for

FSC 4820

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MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification for
MIL-P-7936	Parts and Equipment, Aeronautical, Preparation for Delivery
MIL-L-10547	Liners, Case, and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible
MIL-C-52078	Cap-Plug, Cap, and Plug, Protective, Plastic, Dust and Moisture Seal
MIL-D-70327	Drawings, Engineering and Associated Lists

STANDARDS

Military

MIL-STD-8	Dimensioning and Tolerancing
MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MS33586	Metals, Definition of Dissimilar

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction.- This specification provides for preproduction inspection (see 4.3).

3.2 Data.- Unless otherwise specified in the contract or order, no data are required by this specification or any of the documents referenced in section 2 (see 6.2).

3.3 Materials.- Materials and processes used in the manufacture of low-pressure air relief valves shall conform to applicable Government specifications. Materials and processes conforming to contractor's specifications may be used provided the specifications are released by the procuring activity and contain provisions for adequate tests. The use of contractor's specifications shall not constitute waiver of Government inspection. The use of leather is prohibited.

3.3.1 Metals.- All metals used in the construction of the valve shall be corrosion resistant or shall be suitably protected to resist corrosion. The use of dissimilar metals, as defined by MS33586, shall be avoided whenever practicable. The use of magnesium or any alloy thereof is prohibited.

3.3.2 Nonmetals.- When utilized, nonmetallic parts, such as packings, gaskets, and valve seats, shall conform to MIL-P-5516.

3.3.3 Castings.- Castings, when used, shall be clean, sound, and free from blowholes, porosity, cracks, and other defects.

3.3.4 Selection of materials.- Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143, except as provided in 3.3.4.1 and 3.3.4.2.

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3.3.4.1 Standard parts.- Standard parts (MS and AN) shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part numbers. In the event there is no suitable corresponding standard part in effect on date of invitation for bids, commercial parts may be used, provided they conform to all requirements of this specification.

3.3.4.2 Commercial parts.- Commercial utility parts, such as screws, bolts, nuts, and cotter pins may be used, provided they possess suitable properties and are replaceable by the MS or AN parts without alteration, and provided the corresponding MS or AN part numbers are referenced in the parts list and, if practicable, on the contractor's drawings.

3.4 Design.- The valve shall be so designed as to be capable of being installed in or removed from the receptacle shown on figure 1 by use of standard hexagonal deep socket wrenches. No part of the valve shall be such as could cause interference with installation and removal; or change the pressure relief setting upon installation.

3.5 Construction.- The valve shall be so constructed that it will withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, and service usage.

3.6 Interchangeability.- All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of MIL-D-70327.

3.7 Screw threads.- All screw threads shall be in accordance with MIL-S-7742.

3.7.1 Locking of parts.- All internal and external threaded parts shall be positively locked.

3.7.2 Pipe threads.- Pipe threads shall be in accordance with MIL-P-7105.

3.8 Lubrication.- The valve shall operate satisfactorily without the use of lubricants.

3.9 Performance.-

3.9.1 Static pressure leakage.- The valves shall not leak under a pressure of 5 ±0.1 psi, when tested as specified in 4.6.2.

3.9.2 Dynamic cracking pressure.- Dynamic cracking pressure is the minimum pressure at which air flows through the valve at a rate of approximately 0.1 cfm. The valves shall meet the following requirements when tested as specified in 4.6.3, 4.6.3.1, and 4.6.3.2.

3.9.2.1 Dynamic cracking pressure at room temperature.- Dynamic cracking pressure shall be not more than 11 psi or less than 9 psi.

3.9.2.2 Dynamic cracking pressure at high and low temperature.- Dynamic cracking pressure shall be not more than 18 psi or less than 9 psi, when tested at 160° and -40° F.

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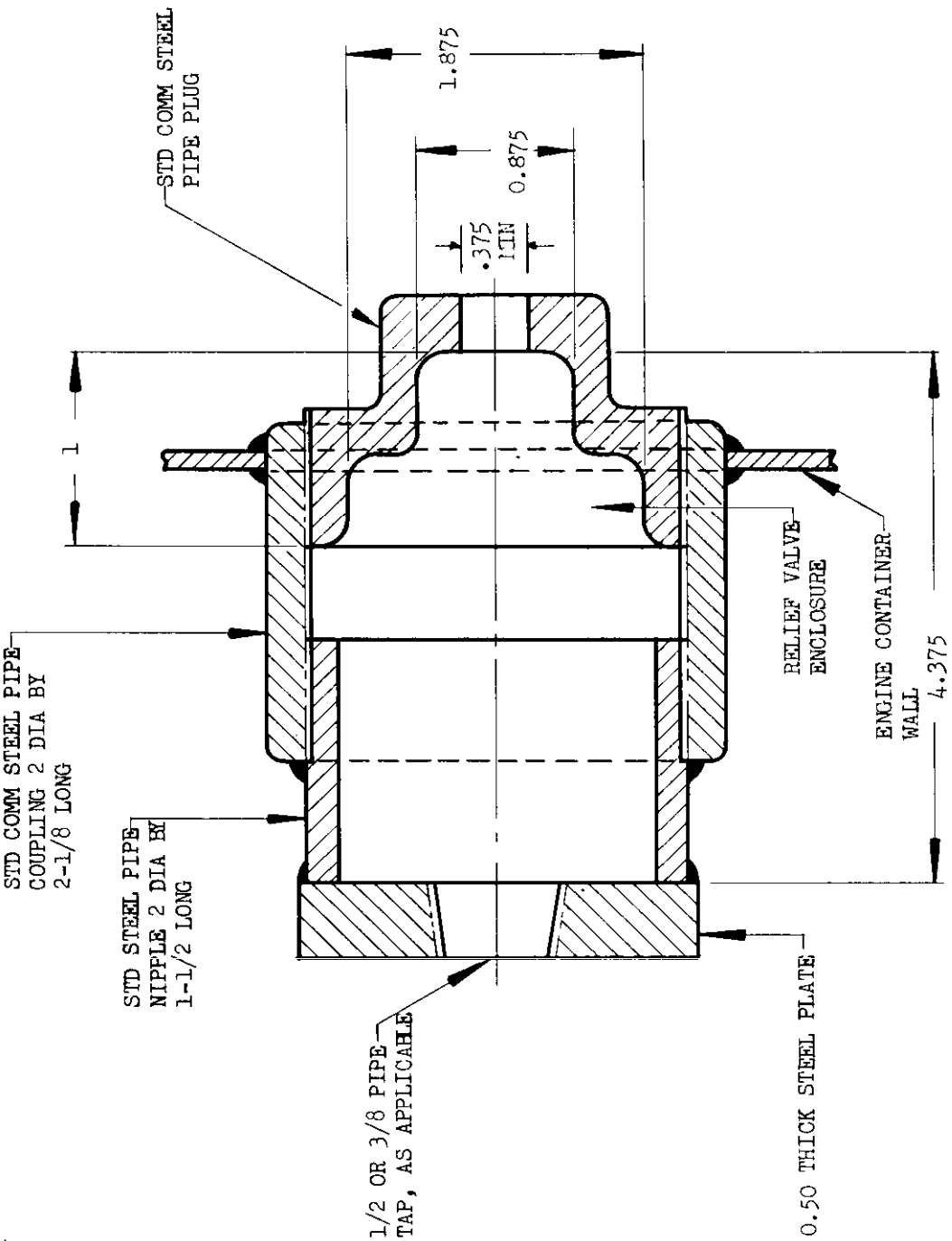


FIGURE 1. Relief valve receptacle

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3.9.3 Flow rate.- The minimum flow of free air through the valves, at 20 ±0.5 psi, shall be 5 cfm.

3.9.4 Reseating pressure.- The reseating pressure shall be not less than 6.5 psi when tested in accordance with 4.6.5.

3.9.5 Rough handling.- The valves shall pass the tests specified in 4.6.2 and 4.6.3 after being subjected to the test specified in 4.6.6.

3.9.6 Accelerated aging.- Initial dynamic cracking pressure (room temperature), after aging as specified in 4.6.7, shall not be more than 18 or less than 9 psi, and, thereafter, not more than 11 or less than 9 psi. The valves shall also pass the test specified in 4.6.2 after the aging period.

3.10 Identification of product.- The valves shall be marked for identification in accordance with MIL-STD-130.

3.11 Workmanship.-

3.11.1 General.- Each valve and all parts thereof shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to machining of mating parts, finish of sealing surfaces, and freedom of parts from burrs and sharp edges.

3.11.2 Dimensions.- Dimensions and tolerances shall be in accordance with MIL-STD-8. Where dimensions and tolerances may affect interchangeability, consistent operation, or performance of the valve, they shall be held or limited accordingly.

3.11.3 Cleaning.- The valve shall be thoroughly cleaned of loose metal chips and other foreign material prior to and after final assembly. Burrs and sharp edges, as well as excess flash on molded parts, shall be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections.- The examining and testing of the valves shall be classified as:

(a) Preproduction inspections (4.3)
(b) Quality conformance inspections (4.4)

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4.3 Preproduction inspections.-

4.3.1 Sampling.- Preproduction test samples shall consist of three valves of each manufacturer's part number upon which approval is desired. When specified, samples shall be accompanied by a complete set of reproducible detail and assembly drawings (see 6.2). Each preproduction test sample shall be plainly identified by securely attached tags, clearly and durably marked with the following information, and shall be forwarded to the testing activity specified by the procuring activity (see 6.2):

Sample for preproduction inspection
Submitted by (name) (date) for preproduction inspections
in accordance with MIL-V-8712A under authorization
(reference letter of authorization)
VALVE; AIR RELIEF, LOW PRESSURE
Size (A or B, as applicable)
Manufacturer's part number
Name of manufacturer

4.3.2 Manufacturer's drawings.- Manufacturer's drawings (see 6.2) shall include a cutaway section showing all parts in their normal assembled position. The following data shall be furnished on or together with the assembly drawing:

(a) Dimensions
(b) Location, size and type of inlet and outlet ports
(c) Materials, treatment and finish (identify each by specification number)
(d) Construction

4.3.3 Inspections.- The preproduction inspection of the valves shall consist of the examination and tests performed on each sample, as specified, conducted in the order listed:

Valve No. 1

(a) Examination of product
(b) Accelerated aging
(c) Static pressure leakage
(d) Dynamic cracking pressure (room temperature)
(e) Flow rate
(f) Reseating pressure

Valve No. 2

(a) Examination of product
(b) Rough handling
(c) Flow rate
(d) Reseating pressure
(e) Static pressure leakage
(f) Dynamic cracking pressure (room temperature)

Valve No. 3

(a) Examination of product
(b) Static pressure leakage
(c) Dynamic cracking pressure (room temperature)
(d) Flow rate
(e) Reseating pressure
(f) Dynamic cracking pressure (high and low temperature)

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4.4 Quality conformance inspections.- Quality conformance inspections shall consist of:

(a) Individual inspections (4.4.1)
(b) Sampling inspections (4.4.2)

4.4.1 Individual inspections.- Each valve shall be subjected to the examination of product (4.6.1).

4.4.2 Sampling inspections.- Samples shall be selected at random in accordance with MIL-STD-105, inspection level II, acceptable quality level 1.5, and shall be subjected to:

(a) Static pressure leakage (4.6.2)
(b) Dynamic cracking pressure (room temperature) (4.6.3)
(c) Flow rate (4.6.4)
(d) Reseating pressure (4.6.5)

4.5 Test conditions.-

4.5.1 Temperature.- Unless otherwise specified, tests shall be conducted at room temperature, 70° to 90° F.

4.5.2 Pressure.- Unless otherwise specified, all pressures specified are gage pressures.

4.6 Inspection methods.-

4.6.1 Examination of product.- Each valve shall be examined to determine conformance to the requirements of this specification, and the manufacturer's drawings, for which there are no tests specified.

4.6.2 Static pressure leakage.- Air, at a pressure of 5 ±0.1 psi, shall be applied to the inlet port of the valve for 1 hour. Any suitable means may be employed for detection of air leakage through the valve.

4.6.3 Dynamic cracking pressure (room temperature).- Air, at a pressure of 5 psi, shall be applied to the inlet port of the valve with the outlet ports open to the atmosphere. The pressure shall be increased in 0.1-psi increments until dynamic cracking occurs. A typical setup is shown on figure 2.

4.6.3.1 Dynamic cracking pressure (high temperature).- The dynamic cracking pressure test shall be conducted as detailed in 4.6.3 and in accordance with the high temperature test Procedure II of MIL-E-5272, except that the conditioning time shall be 16 hours.

4.6.3.2 Dynamic cracking pressure (low temperature).- The dynamic cracking pressure test shall be conducted as detailed in 4.6.3, and in accordance with the low temperature test Procedure I of MIL-E-5272, except that the conditioning temperature shall be -40° F.

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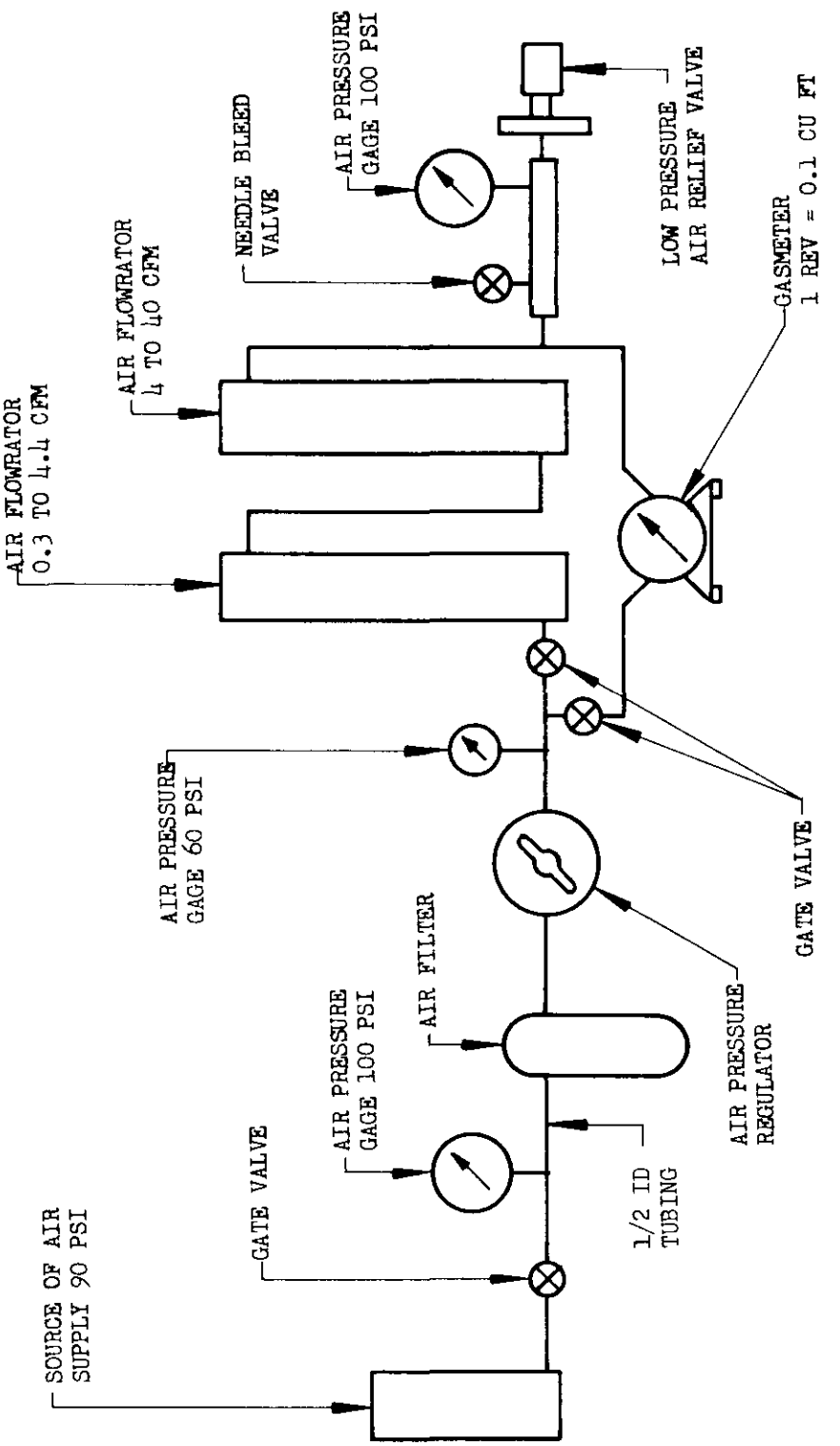


FIGURE 2. Typical test set-up for dynamic cracking pressure and flow rate tests

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4.6.4 Flow rate.- Pressure at 20 ±0.5 psi shall be applied to the inlet port of the valve with the outlet ports open to the atmosphere and the air flow measured. A typical setup is shown on figure 2.

4.6.5 Reseating pressure.- The relief valve shall be operated several times and then the pressure shall be raised to 20 ±0.5 psi. Subsequently, the air flow shall be reduced to a negligible amount and allowed to exhaust into a beaker of water. The pressure shall be reduced gradually until no bubbles appear for a period of 3 minutes. The final pressure is the reseating pressure. A typical test setup is shown in figure 2.

4.6.6 Rough handling.- The valve shall be mounted in a container conforming to figure 3. The container shall be dropped three times on each of three mutually perpendicular sides, for a total of nine drops from a height of 3 feet, upon a concrete surface.

4.6.7 Accelerated aging.- The valve shall be aged for 7 days in an air oven at 160° F and then cooled to room temperature for 24 hours.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation and packaging shall be level A or C, as specified in the contract or order (see 6.2).

5.1.1 Level A.- Unless otherwise specified in the contract or purchase order, each valve shall be prepared for delivery in accordance with MIL-P-7936, and packaged in accordance with Method III of MIL-P-116. Inlet and outlet ports shall be sealed with closures conforming to MIL-C-5501 or MIL-C-52078 to protect the threads and to prevent entrance of foreign matter.

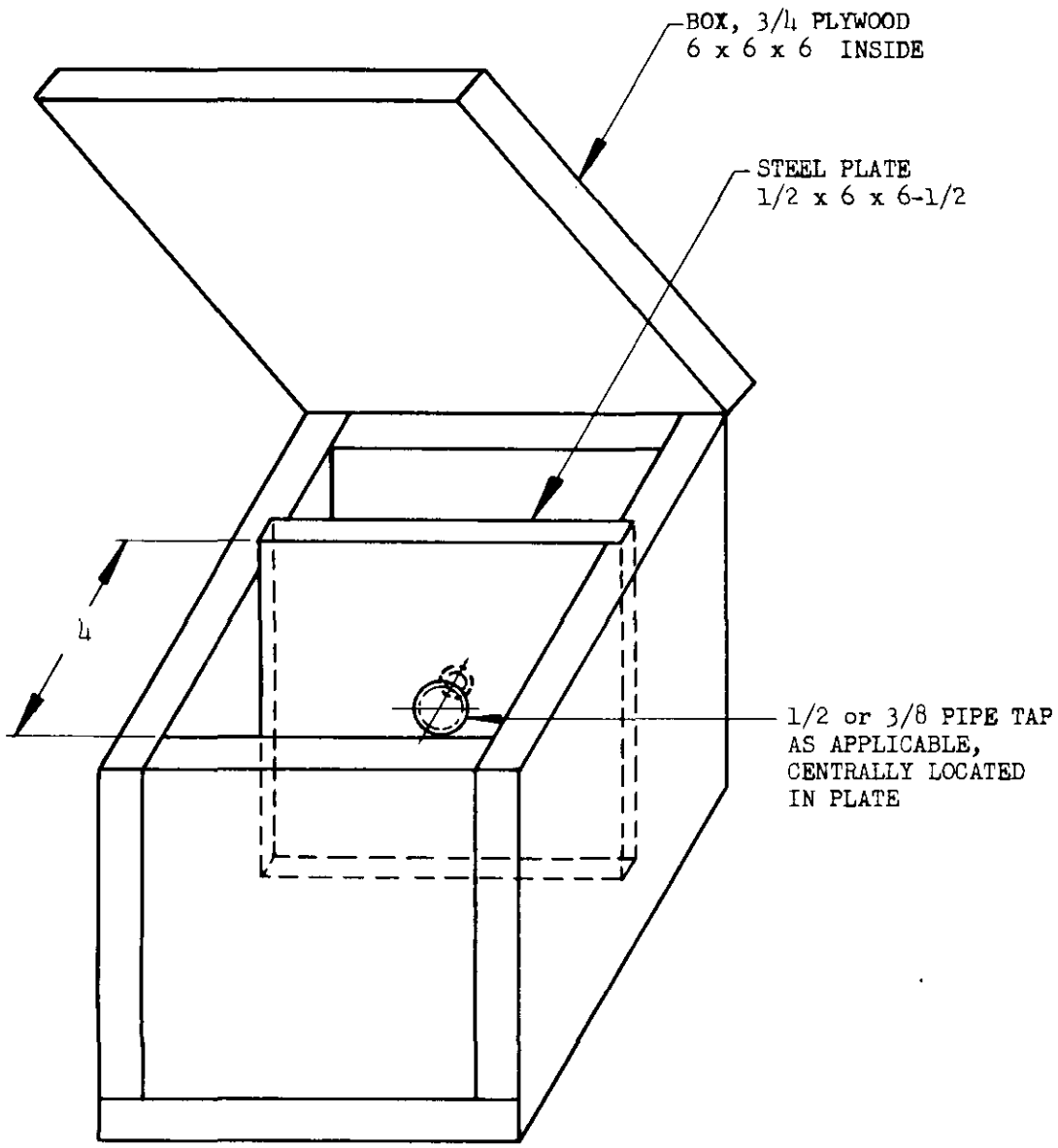
5.1.2 Level C.- Each valve shall be preserved and packaged in accordance with MIL-P-7936.

5.1.3 Intermediate packaging.- Unless otherwise specified, valves packaged as specified in 5.1 shall be packaged in intermediate containers conforming to PPP-B-636.

5.2 Packing.- Unless otherwise specified, packing shall be level A or C, as specified (see 6.2) and shall be in accordance with MIL-P-7936.

5.2.1 Level A.- Intermediate containers shall be packed in boxes conforming to PPP-B-601, PPP-B-621, PPP-B-585, or PPP-B-636. Plywood, when used, shall conform to type II, class 2 of NN-P-515. Containers shall be provided with a case liner conforming to MIL-L-10547 and shall be sealed in accordance with the appendix thereto. The case liner will not be required when the intermediate or exterior container conforms to PPP-B-636, class 2, and is sealed at all joints and seams, including the manufacturer's joint, with tape conforming to PPP-T-60.

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DIMENSIONS IN INCHES

FIGURE 3. Container for rough handling test

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5.2.2 Level C.- The unit or intermediate package shall be packed to ensure carrier acceptance and safe delivery to destination in containers conforming to carrier rules and regulations applicable to the mode of transportation and at the lowest rate.

5.3 Marking.- Interior containers and exterior shipping containers shall be marked in accordance with MIL-STD-129, and shall include:

VALVE; AIR RELIEF, LOW PRESSURE (3/8 or 1/2, as applicable)
INCH PORT
MIL-V-8712A
Manufacturer's part No._____
Federal Stock No._____

6. NOTES

6.1 Intended use.- The relief valves are intended for use with pressurized containers as a protective device against excessive internal air pressure.

6.2 Ordering data.- Procurement documents should specify:

(a) Title, number, and date of this specification.
(b) Size of valve desired (see 1.2).
(c) Data requirements (see 3.2, 4.3.1 and 4.3.2).
(d) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(e) Where the preproduction test samples should be sent and the activity responsible for testing (see 4.3.1).

6.3 Preproduction test samples.- For Navy procurement, the preproduction test samples should be forwarded to Commanding Officer, Naval Air Engineering Center, Aeronautical Materials Laboratory, Philadelphia, Pa. 19112.

Custodians:
Army - MO (ERDL)
Navy - WP
Air Force - (11)

Preparing activity:
Navy - WP

Project No. 4820-0060

Reviewer activities:
Army - MO, MI
Navy - WP
Air Force - (11), (69)

User activities:
Army -
Navy -
Air Force -

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CHIEF, BUREAU OF NAVAL WEAPONS

ENGINEERING DIVISION

ATTN: RREN-5

WASHINGTON, D.C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ Direct Government Contract ☐ Subcontract

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

   ☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.)

SUBMITTED BY (Printed or typed name and activity) DATE

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